

A Mechanism-First Audit of the LHC Black-Hole Safety Debate

Public demo generated from static audit artifacts

Abstract

Scientific controversies are often represented as provenance graphs: who made a claim, which document contains it, and who agrees or disagrees. This is useful for attribution, but it does not test whether the physical mechanism exists. We demonstrate a second graph on the LHC microscopic-black-hole example. The public audit reads static equation witnesses and separates a provenance graph from a mechanism graph. In 492 selected arXiv sources, the strict gate found 1408 equation witnesses, 204 usable formula-clean mechanism nodes, 72 LHC-black-hole case-relevant nodes, and 35 evidence-grade case nodes. It found 0 formula-clean direct LHC-safety mechanisms, 1 collider-threshold or event-selection hook, and 28 astrophysical black-hole analogues. The result is not a final safety review. It is a demonstration that the serious audit layer is mechanism translation from adjacent physics, not claim counting.

1 The Problem

A provenance graph can tell us which paper asserted safety, danger, or a criticism. It cannot by itself test the branch structure of the physical argument. For the LHC black-hole case, the relevant scientific question is not whether a sentence says “safe” or “dangerous”. The question is whether the chain of physical mechanisms can be made explicit:

production $\rightarrow$ evaporation or stability $\rightarrow$ capture/stopping $\rightarrow$ accretion/growth $\rightarrow$ astrophysical bound.

This is why the demo builds two graphs from the same papers.

2 Two Graphs in the Same Archive

Provenance graph. Nodes are sources and claims. Edges record that a source makes a claim or belongs to a claim family. This graph is inspectable but does not decide whether a claimed mechanism closes.

Mechanism graph. Nodes are equation witnesses and local formula transitions. Edges record source-local route transitions and cross-source route analogues. A node is evidence-grade only if the local formula window contains both the black-hole case and a mechanism category. Source-level words alone are not enough.

3 Mechanism Gates

The public demo uses six route labels as a readable interface to equation transformations:

transport/flow	$\partial_t q + \nabla \cdot J = S,$
constraint/closure	$C(q, J, \lambda) = 0,$
spectral/operator	$Lq = \lambda q,$
boundary/weak form	$\int_{\Omega} \phi Lq = \int_{\Omega} \phi f, \quad Bq = b,$
commutator/residual	$[A, B] = AB - BA,$
discrete protocol	$x_{n+1} = \Phi(x_n, u_n).$

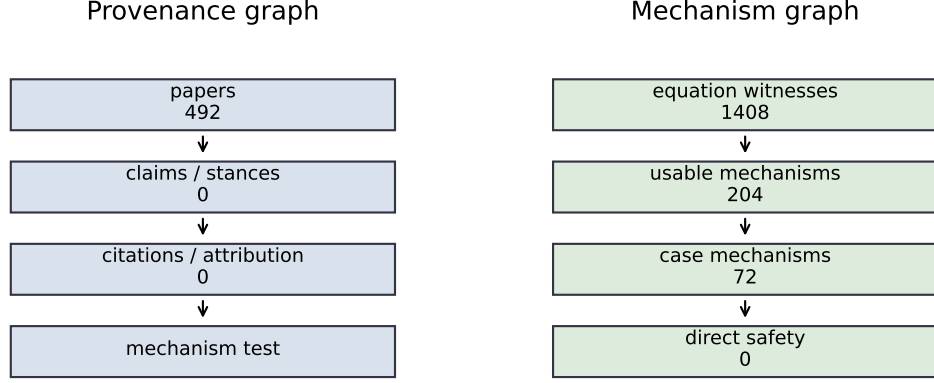


Figure 1: A provenance graph and a mechanism graph answer different questions. The first organizes attribution. The second tests whether equation-level mechanisms exist and can be translated.

These are not final physical claims. They are audit gates that decide which equations are usable as mechanism receipts.

4 Results

The strict run gives the following public counts:

equation witnesses	1408
usable formula-clean mechanism nodes	204
case-relevant nodes	72
evidence-grade case nodes	35
direct LHC-safety mechanisms	0
collider-threshold hooks	1
astrophysical black-hole analogues	28.

The result is precise: the selected corpus does not provide a formula-clean direct LHC-safety mechanism under the current gates. It does provide adjacent black-hole physics: accretion, evaporation, capture, mass growth and compact-object survival. The audit therefore becomes a translation problem.

5 Mechanism Translation to the LHC Branch

The LHC branch needs a production or event-selection condition, for example

$$\sqrt{s} > M_D,$$

followed by a branch decision. If evaporation dominates, one expects a mass-loss condition schematically of the form

$$\frac{dM}{dt} < 0, \quad \tau_{\text{evap}} \ll \tau_{\text{capture}}.$$

If a stable object is assumed, the relevant mechanism changes to capture and growth:

$$\frac{dM}{dt} = \rho \sigma(M) v.$$

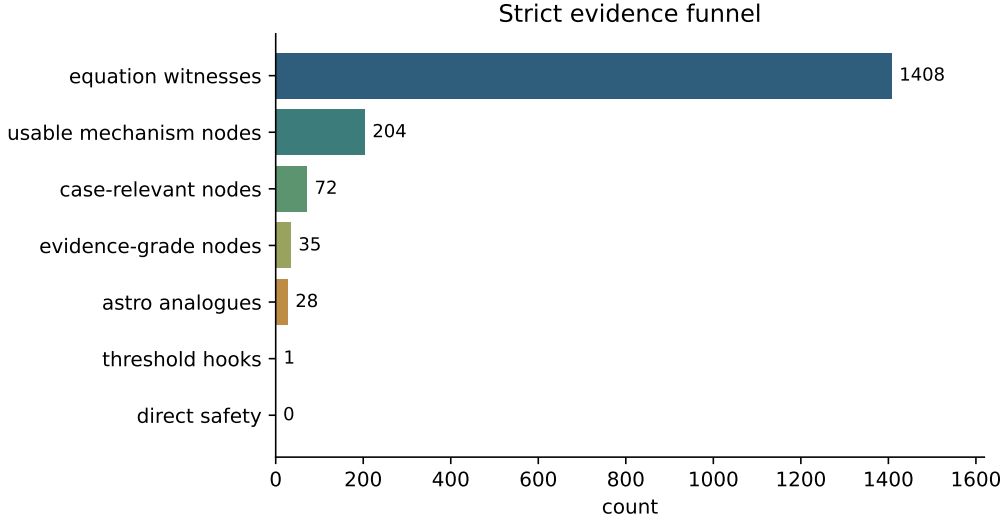


Figure 2: Strict evidence funnel. Most extracted witnesses are rejected or kept only as audit material. The useful public result is the branch split: no direct safety mechanism under the gate, one threshold hook, and many adjacent astrophysical mechanisms.

Astrophysical survival bounds then enter as falsifiers. A dangerous growth branch would have to be compatible with compact-object observations; schematically,

$$t_{\text{grow}} < t_{\text{WD}}$$

would conflict with the observed survival of white dwarfs if the same mechanism were already realized in nature. The public graph found many adjacent astrophysical mechanisms of this type, but no direct formula-clean LHC-safety mechanism under the strict gate.

6 Sparse-Attention Support

The sparse-attention audit reads the static mechanism graph and asks which routes co-activate in the evidence-grade case layer. Its role is interpretive: it supports the branch split; it does not create new fingerprints.

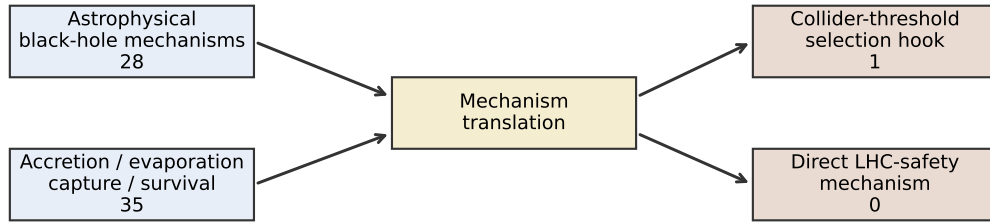
- Sparse-attention receipts are generated when `sparse_attention_audit.json` is available.

7 Why This Beats a Provenance-Only Graph

A provenance-only graph can identify papers, authors, dates and claim families. It is useful for scrutiny of attribution. It fails when the important question is mechanistic: whether the physical branch closes. In this demo, provenance can organize disagreement about LHC safety, but the mechanism graph shows why the scientific evidence is indirect. The direct safety branch is empty under the strict gate. The adjacent astrophysical branch is populated. This distinction is not visible from a claim map alone.

8 Boundary

This report does not claim to resolve the LHC safety question. It demonstrates a public, inspectable audit object built from static outputs. The Hyperion core is not included in this



Result: audit shifts from claim counting to mechanism transfer.

Figure 3: The demo result is a mechanism-translation map. Adjacent black-hole mechanisms provide the inspectable scientific substrate. The collider branch supplies only a threshold hook in this run.

repository. The public files contain derived graph artifacts, run summaries, figures and report text only.

Sparse-attention audit requires
sparse_attention_audit.json

Run scripts/build_sparse_attention_audit.py on the full static graph.

Figure 4: Sparse co-activation between case branches and mechanism routes. The important signal is concentration in adjacent astrophysical mechanisms, not direct collider-safety derivations.